

tensortools: Tools for Tensor-Variate Distributions and Decompositions

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Motivation

- Many datasets are naturally high-dimensional with structured covariance
 - RGB video - order 4 (x, y, color channel, time)
 - Medical imaging - order 4 (x, y, z, time)
 - Sensors over space and time

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 - RGB video - order 4 (x, y, color channel, time)
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 - Sensors over space and time
- Established software implementations and methods exist for vector- and matrix-variate data
- Analysts often matricize tensor data to use these methods

Tensors

- Tensors - generalization of vectors and matrices (Kolda and Bader 2009)
- Order - number of indices needed to specify a value
 - Vector has order 1, matrix has order 2
- Mode - specific axis of tensor (rows = 1st mode, cols = 2nd mode)

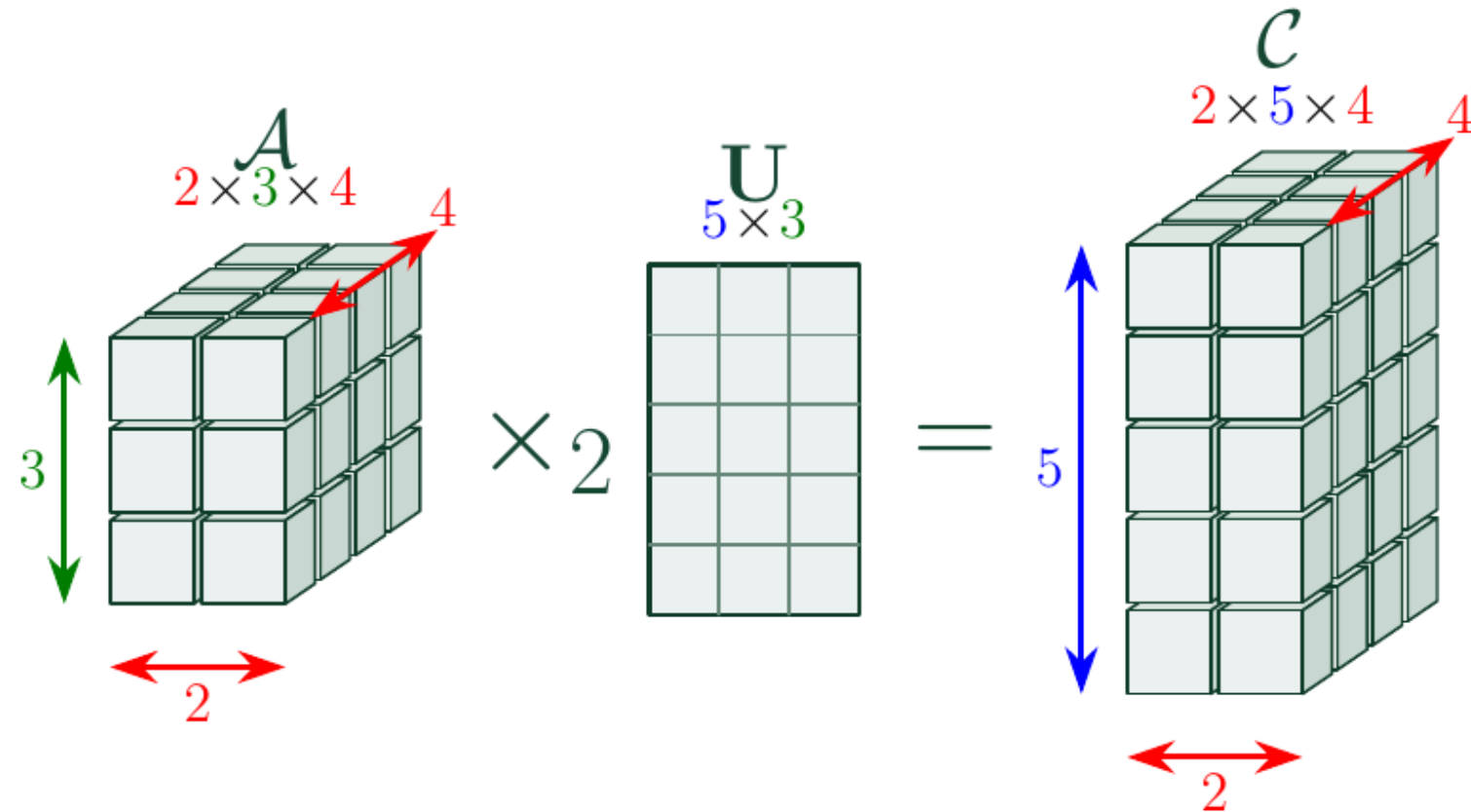
$\times_n$ Product

- Let $\mathcal{A} \in \mathbb{K}^{I_1 \times I_2 \times \cdots \times I_n \times \cdots \times I_O}$ and a matrix $\mathbf{U} \in \mathbb{K}^{J \times I_n}$
- Then $\mathcal{C} = \mathcal{A} \times_n \mathbf{U}$ has entries

$$c_{i_1, \dots, i_{n-1}, j, i_{n+1}, \dots, i_O} = \sum_{k=1}^{I_n} a_{i_1, \dots, i_{n-1}, k, i_{n+1}, \dots, i_O} u_{j, k}$$

- The result has dimensions $I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_O$ and remains order O
- Acts along mode n while leaving the other modes intact

$\times_n$ Product



- Analogous to matrix multiplication
- The matrix acts along one mode of the tensor, forming weighted combinations along that mode

Tensor Variate Normal (TVN)

Object	Standard-normal	Transformation	Covariance structure
Scalar	$Z \in \mathbb{R}^1$	$X = \mu + \sigma Z$	σ^2

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Tensor	$\mathcal{Z} \in \mathbb{R}^{I_1 \times \cdots \times I_D}$	$\mathcal{X} = \mathcal{M} + \mathcal{Z} \times_{d=1}^D \mathbf{A}_d$	$\Sigma_d = \mathbf{A}_d \mathbf{A}_d^T$

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$$\mathbf{X} = \mathbf{M} + \mathbf{Z} \times_1 \mathbf{A} \times_2 \mathbf{B}$$

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- Start with IID standard normal draws in the structure, apply a linear transformation, add a mean (Ohlson et al. 2013)

Tensor Variate Normal (TVN)

- Model assumption: separable covariance structure
 - Each mode d has a covariance factor Σ_d
 - Together, the mode-specific factors define the full covariance:

$$\text{vec}(\mathcal{X}) \sim N(\text{vec}(\mathcal{M}), \Sigma_D \otimes \cdots \otimes \Sigma_1)$$

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- Scaling identifiability issue: covariance structure is the same for different scalings of the covariance matrices
 - $\Sigma_2 \otimes \Sigma_1 = a\Sigma_2 \otimes \frac{1}{a}\Sigma_1$
 - Only the full Kronecker product is uniquely determined (Manceur and Dutilleul 2013)

TVN in tensortools

- `tensortools` supports `r` and `d` functions similar to base R
- `rtnorm()` requires arguments `n`, `mu`, `sigmas`
 - `mu` is the mean tensor. Its dimension is the same as the draws $\mathcal{X}$
 - `sigmas` expects a list of covariance matrices. For mode d , $\Sigma_d \in \mathbb{R}^{I_d \times I_d}$

```

1 (draws <- rtnorm(n = 100, mu = mu_true, sigmas = list(S1, S2, S3)))
2 # <tensor[100]>
3 # 100 draws of shape 2 × 3 × 4
4 #
5 # # Preview: 3 draws × 6 indexed entries
6 #           [1,1,1] [2,1,1] [1,2,1] [2,2,1] [1,3,1] [2,3,1]
7 # draw 1 1.933572 3.758023 2.866924 5.553246 5.084797 7.007887
8 # draw 2 1.458150 2.540761 4.475832 3.368194 4.383154 5.348898
9 # draw 3 2.041212 4.245815 3.160716 5.029540 5.223698 7.717329
10 #
11 # # ... 97 more draws and 18 more entries per draw

```

TVN in tensortools

- Density evaluation is also supported with `dtnorm()`
 - Requires arguments `x`, `mu`, `sigmas`

```
1 dtnorm(draws[1], mu_true, sigmas = list(S1, S2, S3), log = TRUE)
2 # [1] -32.97781
```


TVN in tensortools

- MLE estimation is supported with `tensor_mle()`
 - Requires arguments `draws` and `model`
 - Output is a list of the MLEs along with the log-likelihood, k, AIC, BIC

```

1 tvn_mles <- tensor_mle(draws, model = "normal")
2
3 tvn_mles$sigmas[[1]]
4 #           [,1]      [,2]
5 # [1,] 1.0236879 0.4828938
6 # [2,] 0.4828938 0.9763121
7 S1
8 #           [,1] [,2]
9 # [1,] 1.0 0.5
10 # [2,] 0.5 1.0

```

Tensor class

- Tensor class is internally an array
 - Stores n IID observations, operations apply to each one

```
1 print(draws[1:2], entries = 3)
2 # <tensor[2]>
3 # 2 draws of shape 2 × 3 × 4
4 #
5 # # Preview: 2 draws × 3 indexed entries
6 #      [1,1,1] [2,1,1] [1,2,1]
7 # draw 1 1.933572 3.758023 2.866924
8 # draw 2 1.458150 2.540761 4.475832
9 #
10 # # ... 21 more entries per draw
```

Tensor class

```

1 centered_draws <- draws - mu_true
2
3 print(centered_draws[1:2], entries = 3)
4 # <tensor[2]>
5 # 2 draws of shape 2 × 3 × 4
6 #
7 # # Preview: 2 draws × 3 indexed entries
8 #           [1,1,1] [2,1,1] [1,2,1]
9 # draw 1 0.9335718 1.758023 -0.1330757
10 # draw 2 0.4581498 0.540761 1.4758318
11 #
12 # # ... 21 more entries per draw
13
14 print(n_prod(centered_draws[1:2], 1, S1), entries = 3)
15 # <tensor[2]>
16 # 2 draws of shape 2 × 3 × 4
17 #
18 # # Preview: 2 draws × 3 indexed entries
19 #           [1,1,1] [2,1,1] [1,2,1]
20 # draw 1 1.8125833 2.2248089 0.6435474
21 # draw 2 0.7285303 0.7698359 1.1599286
22 #
23 # # ... 21 more entries per draw

```

Application: GRABMyo

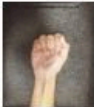
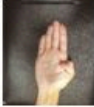
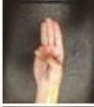
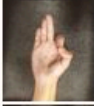
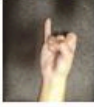
- GRABMyo (Gesture Recognition and Biometrics Electromyogram) data from **43 participants** treated as IID observations ([Jiang et al. 2024](#); [Pradhan et al. 2022](#))
- Each participant performed **16 hand gestures** in 7 repeated trials
- There were **28 active EMG channels**
- Collapse each 5-second recording at 2,048 Hz into **16 time bins**
- Average the seven trials for each participant and gesture after time binning

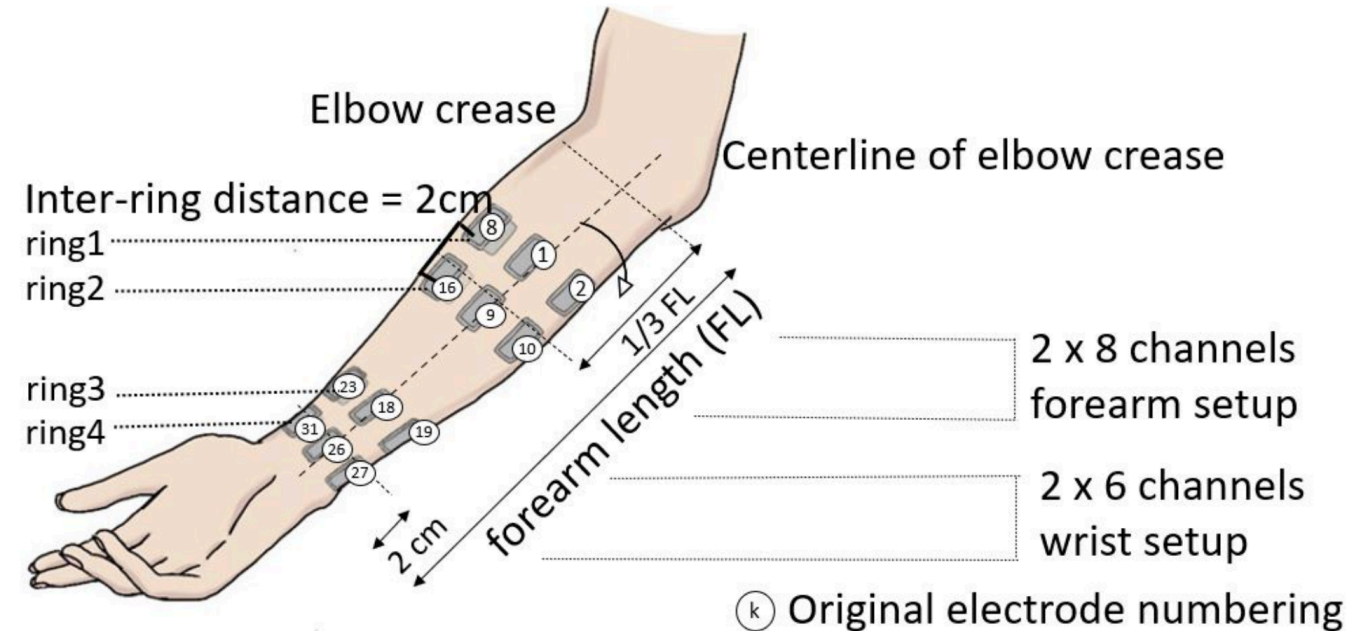
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- Average the seven trials for each participant and gesture after time binning
- Each observation is a **16 gestures $\times$ 28 channels $\times$ 16 time bins** tensor

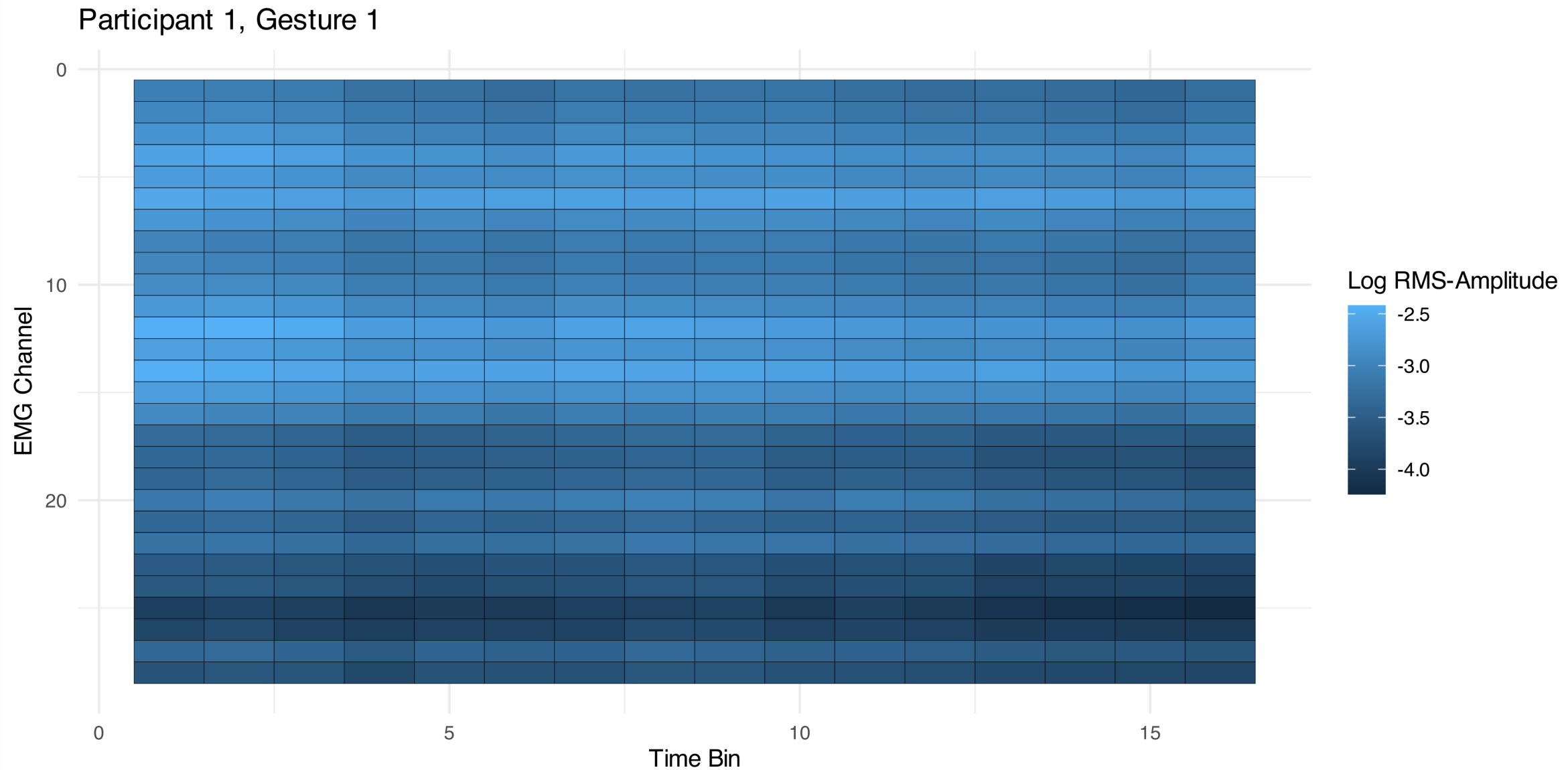
GRABMyo: electrode layout and gesture set

TABLE
GESTURE LIST

Gesture	Description	Gesture	Description
	Lateral prehension (LP)		Index finger extension (IFE)
	Thumb adduction (TA)		Thumb extension (TE)
	Thumb and little finger opposition (TLFO)		Wrist flexion (WF)
	Thumb and index finger opposition (TIFO)		Wrist extension (WE)
	Thumb and little finger extension (TLFE)		Forearm pronation (FP)
	Thumb and index finger extension (TIFE)		Forearm supination (FS)
	Index and middle finger extension (IMFE)		Hand open (HO)
	Little finger extension (LFE)		Hand close (HC)



GRABMyo Values



GRABMyo Model Output

Tensor-normal and matrix-normal fits to the GRABMyo data.

model	dimensions	loglik	k	BIC	AIC	secs
tensor	16 x 28 x 16	637062.9	7844	-1244623	-1258438	1.5
mat2	28 x 256	676266.7	40469	-1200321	-1271595	3.2
mat3	16 x 448	737814.6	107879	-1069875	-1259871	15.0
mat1	16 x 448	722443.7	107879	-1039133	-1229129	6.7

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- Matrix-normal models have larger likelihoods because they are less constrained
- BIC favors the tensor model: reduction in complexity is worth the loss in likelihood

Skewed Tensor Distributions

$$\mathcal{X} = \mathcal{M} + W\mathcal{A} + \sqrt{W}\mathcal{V}, \quad \mathcal{V} \sim \text{TVN}(\mathbf{0}, \{\Sigma_d\}_{d=1}^D)$$

(Barndorff-Nielsen et al. 1982)

Distribution	Distribution of W	Param	Support
Skewed-t	Inverse-gamma($v/2$, $v/2$)	v	$v > 0$
Generalized hyperbolic	Gen Inv Gauss(ω , 1 , λ)	λ , ω	$\omega > 0$; $\lambda \in \mathbb{R}$
Variance-gamma	Gamma(γ , γ)	γ	$\gamma > 0$
Normal inverse Gaussian	Inv Gauss(1 , κ)	κ	$\kappa > 0$

Skewed Tensor Distributions in tensortools

- `tensortools` supports `r`, `d`, and MLE functions for all of these distributions
- MLE estimation is done using the EM algorithm

```

1 (skewt_draws <- rtskewt(n = 1e3, mu = mu_true, sigmas = sigmas_true,
2                       skew = skew_true, nu = 20))
3 # <tensor[1,000]>
4 # 1,000 draws of shape 2 × 3 × 4
5 #
6 # # Preview: 3 draws × 6 indexed entries
7 #      [1,1,1] [2,1,1] [1,2,1] [2,2,1] [1,3,1] [2,3,1]
8 # draw 1 3.643947 4.425661 5.973282 9.331688 8.096839 10.628325
9 # draw 2 2.994855 3.973106 3.989750 5.064052 7.983755 9.506291
10 # draw 3 2.052635 2.889480 4.742515 6.128241 7.228344 10.330539
11 #
12 # # ... 997 more draws and 18 more entries per draw
13
14 mle_skewt <- tensor_mle(skewt_draws, model = "skewt")

```

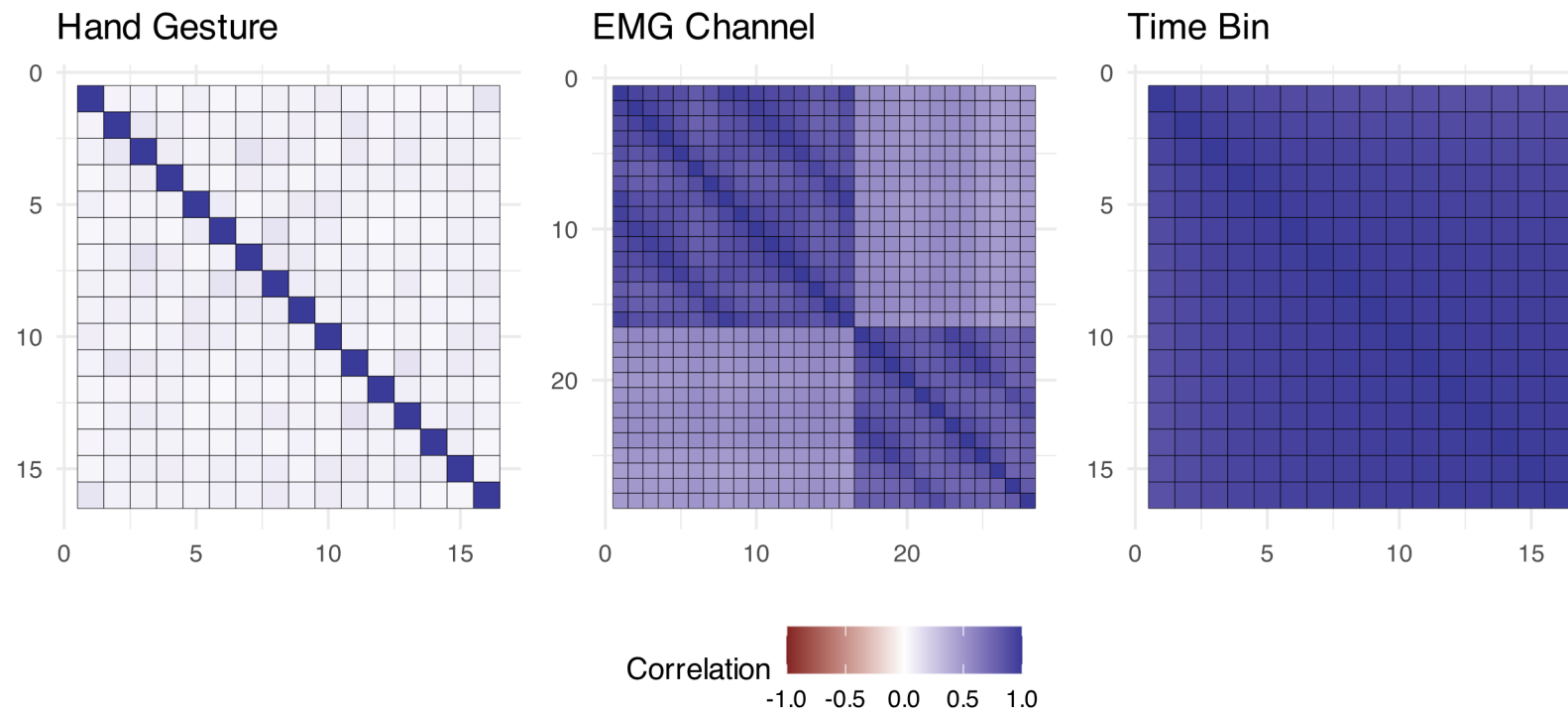
GRABMyo Data with Skew

Tensor distribution fits to the 43 processed GRABMyo participant tensors.

model	loglik	k	AIC	BIC	secs
Skew-t	651376.6	15013	-1272727	-1246286	6.4
Gen Hyper	651375.5	15014	-1272723	-1246280	6.4
Norm Inv Gauss	651362.5	15013	-1272699	-1246258	6.9
Var Gamma	651260.9	15013	-1272496	-1246055	5.3
Normal	637062.9	7844	-1258438	-1244623	1.5

- Skew- t has lowest AIC and BIC even with an additional skew tensor
- AIC and BIC favor skew- t over normal, consistent with skewness

Interpretation



- Cross-gesture correlations are generally weak while time-bin correlations are strong
- EMG has two blocks of stronger within-group correlation: 16 forearm channels and 12 wrist channels

Thank You!

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- Jiang, Ning, Ashirbad Pradhan, and Jiayuan He. 2024. "Gesture Recognition and Biometrics ElectroMyogram (GRABMyo)." *PhysioNet*, ahead of print, June.
- Kolda, Tamara G., and Brett W. Bader. 2009. "Tensor Decompositions and Applications." *SIAM Review* 51 (3): 455–500.
- Manceur, Ameer M., and Pierre Dutilleul. 2013. "Maximum Likelihood Estimation for the Tensor Normal Distribution: Algorithm, Minimum Sample Size, and Empirical Bias and Dispersion." *Journal of Computational and Applied Mathematics* 239: 37–49.
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- Pradhan, Ashirbad, Jiayuan He, and Ning Jiang. 2022. "Multi-Day Dataset of Forearm and Wrist Electromyogram for Hand Gesture Recognition and Biometrics." *Scientific Data* 9: 733.

- Thank you to my advisors Dr. David Kahle and Dr. Michael Gallagher
- Any questions?



W Latent Distributions

$W \sim \text{Inverse-Gamma}(\nu/2, \nu/2)$

$$f(w|\nu) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} w^{-(\nu/2+1)} \exp\left[-\frac{\nu/2}{w}\right]$$

▪ $W \sim \text{Gamma}(\gamma, \gamma)$

$$f(w|\gamma) = \frac{1}{\gamma^\gamma \Gamma(\gamma)} w^{\gamma-1} e^{-w/\gamma}$$

▪ $W \sim \text{Inverse-Gaussian}(1, \kappa)$

$$f(w|\kappa) = \frac{1}{\sqrt{2\pi}} w^{-\frac{3}{2}} \exp\left\{-\frac{1}{2} \left(\frac{1}{w} + \kappa^2 w\right)\right\}$$

▪ $Y \sim \text{Generalized Inverse Gaussian}(\omega, 1, \lambda)$

$$f(w|\omega, b, \lambda) = \frac{(\omega)^{\frac{\lambda}{2}} y^{\lambda-1}}{2K_\lambda(\sqrt{\omega})} \exp\left\{-\frac{x}{2} \left(w + \frac{1}{w}\right)\right\}$$